ISE549

Management of Innovation and Technology

Week 8 Lecture – Complete Transcript

Strategies for Technology Innovation Firms,
Competitor Dynamics & Innovator's Dilemma

Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

This document is a cleaned and structured full transcript of the Week 8 lectures (ISE549_W08_Lecture.m4a and ISE549_W08_Lecture2.m4a), aligned with the course handout and extension materials.

Note: *Minor transcription corrections have been made for clarity (e.g., "incumbents", place names, technical terms). The content remains faithful to the original spoken lecture.*

1. Lecture Overview & Context

This is Week 8 of ISE549. The session continues the discussion from the previous week and forms the climax of the subject. Students are expected to apply the theories, principles and strategies learned over the past six to seven weeks to real competitive situations between technology innovation firms.

Key topics covered include the technology lifecycle, S-curve, Innovator's Dilemma, ambidextrous organisation, path dependency & evolutionary traps, and competitive dynamics. Several scenarios are provided for practice. Towards the end of the session the format and approach for the Week 12 test are also discussed.

Students are reminded that attendance in person is expected; random checks will be conducted. There are no marks for attendance, but missing the enriched classroom discussion puts students at a competitive disadvantage.

2. The Strategic Landscape: Incumbents vs Challengers

The lecture begins by establishing terminology. An **incumbent** is the firm (or firms) already established in the market, occupying an advantageous position with greater resources, customer base, brand trust and ability to scan the horizon. A **challenger** (or new entrant) is any firm seeking to gain market share at the expense of the incumbent(s). A challenger does not have to be a brand-new company; it can be an existing firm using either existing or new technology.

The ultimate goal of a challenger is not necessarily to "kill" the giant, but to survive, succeed and sustain success in the market. If the incumbent loses ground rapidly, it may eventually exit or be sold off.

2.1 Example: Electric Vehicles

Globally, BYD (China) has the highest production volume (roughly three to four times that of Tesla), while Tesla enjoys broader geographic spread. Both are incumbents in the pure-electric segment. Challengers include:

- Smaller pure-electric vehicle startups that typically attack the lower end of the market (closer to Christensen-style disruptive innovation).
- Traditional petrol/diesel vehicle manufacturers (often forming alliances) that possess large resources, existing customer bases and decades of experience, but suffer from limited EV-specific expertise, slower decision-making due to size, and heavy dependence on existing revenue streams.

2.2 Example: Cloud Computing (AWS)

Amazon Web Services remains the dominant incumbent. Microsoft Azure has closed the gap but has not overtaken AWS. Google Cloud and Oracle are growing but still lag significantly. Later in the lecture the reasons for AWS's sustained success are examined.

3. Competitive Dynamics

Competitive dynamics concerns the actions and responses taken by all firms – both incumbents and challengers. Analysis must be conducted from both perspectives: what the incumbent should do and how the challenger can counteract (and vice versa).

The process begins with competitive analysis: understanding the market and customer needs, locating the firm's product on the S-curve, monitoring emerging technologies, assessing competitors' progress in expertise and market position, and ultimately seeking sustainable market-share gains.

Possible end-states of rivalry include: (1) challengers overwhelm and topple the incumbent; (2) the incumbent crushes the challengers; or (3) prolonged “zombie” or stalemate battles whose outcome is still undecided (a snapshot in time).

4. Rogers’ Theory of Innovation Diffusion

Diffusion refers to the spread of an innovation such that more people become aware of it, accept it, purchase it and adopt it. Rogers’ model identifies five stages that an individual or organisation passes through:

1. **Knowledge / Awareness** – the innovation must first be known.
2. **Persuasion** – the potential adopter must be convinced of its value.
3. **Decision** – the adopter decides to buy or reject.
4. **Implementation** – the innovation is put into use.
5. **Confirmation** – the adopter seeks reinforcement and continues (or abandons) use.

Factors influencing the speed and likelihood of adoption include relative advantage, compatibility, complexity, trialability, observability, customer preferences, complementary assets, competitive threat, government policy/subsidies, reputation (which reduces uncertainty), and entry/exit barriers.

4.1 Categories of Adopters (the classic bell curve)

• **Innovators** – the first to enter; enjoy temporary monopoly but bear the highest technology, market, regulatory and financial risk.

• **Early Adopters** – watch the innovators, learn from their mistakes, and often capture a larger ultimate market share.

• **Early Majority** (~34 %) – enter when suppliers are already established; must compete against rising giants.

• **Late Majority** (~34 %) – enter when the market is approaching saturation or even decline; can exploit underserved niches, newer/cheaper technology generations, and gaps left by earlier players.

• **Laggards** – the very last; success is still possible if strategy is superior (Jensen Huang / NVIDIA is cited as an example of a successful late entrant).

Empirical observation repeatedly shows that early leaders (especially early adopters rather than pure innovators) often achieve roughly three times the market share of the original pioneers. Late entrants can succeed by operating on a superior technological platform that is both more powerful and cheaper.

5. First-Mover Advantages and Disadvantages

Advantages

• Brand loyalty and ability to shape customer preferences.

• Opportunity to establish a dominant design / architecture that locks in customers and forces competitors to comply.

• Control of scarce assets and the ability to erect entry barriers.

• Temporary monopoly while competitors are absent.

Disadvantages / Risks

• Highest technological, market, regulatory and financial risk; deep pockets and patience are required.

- Immature supply chains and enabling technologies.
- Technology ages from day one; continuous investment is needed.
- Lack of complementary products/accessories early on reduces appeal.

Classic illustration of dominant design lock-in: the QWERTY keyboard. Even though research later showed that the Dvorak layout is more ergonomic, the installed base of trained typists made switching prohibitively expensive. The inferior design persists decades later.

6. How Incumbents Successfully Crush Challengers

Cases examined include Netflix (versus earlier rivals), Johnson & Johnson, and AWS. Common successful tactics of incumbents include:

- **Fast-follower strategy** – let the challenger take the initial risk, then rapidly copy (legally and ethically) and launch into a much larger existing user base.
- **Network effects** – the incumbent's larger, more trusted network creates stronger positive feedback than the challenger's.
- **Ecosystem bundling and data advantage** – incumbents possess years of multi-channel customer data that can be analysed for personalisation, prediction and recommendation.
- **Acquisition** – buy the challenger before it becomes large enough to threaten the core business (Microsoft is frequently cited).
- **Supply-chain / resource control** – lock up scarce, valuable resources so that competitors face high entry barriers.

7. Deep Dive: The Netflix Story (Rise, Maturity and Current Trouble)

Netflix entered the market around 1997–1998 as a challenger to Blockbuster (physical DVD rental). Within a few years it had toppled the incumbent. It IPO'd, reached 100 million subscribers by 2017, and enjoyed strong growth for roughly 13 years (≈2007–2020). Competitors such as YouTube Premium, Amazon Prime and Apple TV+ appeared, but Netflix remained the leader for a long period.

By the mid-2020s Netflix is showing clear signs of maturity and stress: large-scale lay-offs (≈1 000 staff / 6 %), stock decline of nearly 42 % over the preceding year and 25 % in 2026, subscription saturation, password-sharing crackdown costs, heavy content and sports investment with uneven returns, and thinning advertising revenue. The firm's trajectory now resembles the classic S-curve plateau followed by decline.

Key lessons: even a highly successful challenger that becomes the new incumbent eventually faces the same maturity pressures, competitive intensity and the need for continuous reinvention.

8. Empirical Pattern: Who Usually Wins?

Across multiple tables compiled from textbooks covering the last 25–35 years (Blockbuster vs Netflix, Nokia/BlackBerry vs Apple, taxi fleets vs Uber, Microsoft vs Slack, Netflix vs Disney+, etc.), late entrants / challengers ultimately prevail in the majority of cases (roughly three-quarters). Only about one-quarter of first-movers retain long-term leadership; a few contests end in draws.

Two primary reasons for incumbent failure are emphasised:

1. **Aging technology platform** – the performance gap with newer technology becomes irreparable.

2. **Management structure and decision-making** – larger organisations suffer more miscommunication, more decision-makers, longer decision cycles and higher risk of evolutionary traps.

Supporting statistic: according to research cited in the lecture, nearly 70 % of incumbent firms lack a defined process for adopting emerging disruptive technologies. They keep executing the current model until it is too late.

Jensen Huang (NVIDIA) is quoted: “It is never too late. If you are first in line you need a good strategy; if you are last in line you also need a good strategy.” Success depends on continuous environmental scanning, learning from predecessors’ mistakes, and sound strategic choices rather than entry timing alone.

9. Ongoing Battle Snapshot: Electric Vehicles (as of late 2026)

China already accounts for a large share of global EV production (roughly four to five times Tesla’s volume). Tesla retains stronger global brand and geographic balance. Traditional ICE manufacturers still command the majority of the installed base worldwide. Regional patterns differ: China has high EV penetration; Europe is a contested battleground for both BYD and Tesla; parts of Asia, South America, Africa and the Middle East remain more resistant because of infrastructure, cost and (in the Middle East) resource considerations. The outcome is still open; continuous monitoring is required.

10. Summary Frameworks for Incumbent / Challenger Strategy

The lecture consolidates the key frameworks students should apply when analysing any competitive situation. They are grouped under four headings:

Marketing & Industry Structure

- 4Ps of marketing (Product, Price, Promotion, Place/Distribution) – intelligent combinations can create differentiation.
- Entry and exit barriers – used both to trap customers and to deter competitors.
- Porter’s Five Forces and PESTLE – external and industry analysis.

Resources & Capabilities

- VRIO / resource-based view – control of rare, valuable, inimitable resources (including rare earths, proprietary data, loyal talent).
- Knowledge management – knowledge is infinitely renewable.
- Dynamic capabilities – especially valuable when internal resources are insufficient; rely on trusted external networks and ecosystems.
- Absorptive capacity – ability to sense, assimilate and exploit external knowledge.

Innovation & Strategy

- Ambidextrous organisation – simultaneously exploit existing technology and explore new technology (“one foot in each boat”).
- Disruptive vs sustaining innovation; open innovation; dominant architecture / design.
- Network effects / network externalities.
- Resource dependence and strategic alliances.

Strategic Dynamics

- S-curve awareness and the need to jump to the next curve before the current one declines.
- Path dependency and evolutionary traps.
- Scenario planning and continuous environmental scanning.

11. Worked Scenario: CRM Software – Comfortable Giant vs AI-Native Challenger

The global CRM market is dominated by two large incumbents controlling ≈75 % of the enterprise segment. Their platforms are deeply embedded and trusted, yet customers complain of complexity, long implementation times and high licensing costs. SMEs remain underserved. Incumbent products are in late maturity (incremental improvements only). Venture-backed AI-native CRM startups offer automatic report generation, workflow configuration and natural-language interaction – a potential performance leap – but reliability and security concerns remain for large enterprises. Regulations on data privacy and AI governance are still uncertain.

11.1 Guidance for the Incumbent

There is no single “correct” answer; the quality of reasoning and explicit assumptions matter most. Suggested lines of thought include:

- **Defend vs cannibalise**: Continue harvesting the profitable existing licence model while simultaneously investing in a separate (or modular) subscription / pay-as-you-go cloud offering aimed at SMEs and more agile enterprises. An ambidextrous structure allows customers to move flexibly between the two models.
- **Sustaining vs disruptive AI**: Assess the maturity timeline of AI, the incumbent’s own absorptive capacity and learning curve, and the challengers’ progress. A pure sustaining path risks being leapfrogged; a pure disruptive path risks destroying current cash-flow. Parallel investment is usually wiser.
- **Acquire, partner or compete**: If challengers are still small, acquisition can bring both technology and talent. If they are already scaling rapidly, partnership may give the incumbent access to the AI engine while giving the challenger access to the installed base – a potential win-win if revenue-sharing is structured carefully.
- **Switching costs & ecosystem**: Leverage long-term contracts, data lock-in, integration depth and complementary services to raise exit barriers while selectively lowering them for migration to the new AI offering.

11.2 Guidance for the Challenger

- **Target SMEs first or go after enterprises?** SMEs are more price-sensitive, less locked-in and more willing to accept a simpler, lower-cost, pay-as-you-go solution. Winning the SME segment builds volume, data and credibility that can later support an enterprise push. Direct enterprise attack faces higher credibility and security hurdles.
- Compete primarily on simplicity, speed of implementation and transparent pricing rather than trying to match every feature of the incumbents.
- Overcome credibility barriers through independent security audits, phased roll-outs, strong reference customers, and possibly white-label or OEM partnerships with trusted players.
- Alliances with system integrators, cloud hyperscalers, or complementary SaaS vendors can accelerate adoption and fill capability gaps.

12. Additional Practice Scenarios (Brief Outlines)

Scenario 2 – Electric Aircraft & Regional Aviation

Three major manufacturers dominate regional aviation (>85 % share). Existing aircraft are highly optimised but approaching the upper limit of the current S-curve. Electric / hybrid-electric challengers promise lower operating cost and emissions, yet face battery and certification hurdles. Governments are beginning to

offer incentives; airports are investing in charging infrastructure. Key questions revolve around protecting existing programmes versus heavy investment in electric technology, use of certification expertise as a barrier, focus on underserved short routes, and alliances with airports and governments.

Scenario 3 – Healthcare Diagnostics Revolution

A few multinationals control ≈80 % of the premium hospital diagnostic market. Machines are accurate but expensive and bulky. Handheld AI-enabled devices offer much lower cost and greater accessibility, currently with lower accuracy that is improving rapidly. Classic Innovator's Dilemma setting: defend the premium segment, launch a low-end disruptive offering, or create a separate business unit. Challengers should prioritise clinics, NGOs and government programmes and emphasise accessibility.

Scenario 4 – Smart Agriculture Ecosystem Battle

Two multinationals dominate large-scale mechanised farming. Small and medium farms find the systems expensive and hard to integrate. Challengers offer autonomous drones, AI monitoring and robotics tailored to smaller farms, but the technologies are fragmented. Data interoperability and ecosystem standards are becoming decisive. Incumbents must decide between open platforms and proprietary lock-in; challengers must decide whether to push a new dominant standard or focus on niches and partnerships with cooperatives and governments.

13. Closing Remarks from the Lecture

There is no absolute right or wrong answer to the scenario questions. What is assessed is the quality of reasoning, the explicit statement of assumptions, and the coherent application of the frameworks taught in the course. Students should always list the assumptions on which their recommendations rest, because the information given in any scenario is inevitably incomplete.

The competitive landscape is dynamic. Today's successful challenger becomes tomorrow's incumbent and will face the same evolutionary pressures. Continuous scanning of technology trajectories, customer pain points, regulatory shifts and competitor moves is essential. Ambidexterity – the disciplined ability to exploit the present while exploring the future – remains one of the most powerful organisational capabilities a firm can develop.

— End of complete lecture transcript —

Source materials: ISE549_W08_Lecture.m4a, ISE549_W08_Lecture2.m4a, ISE549_W08_Handout.pdf, ISE549_W08_Extend.pdf.
Prepared for study purposes. All conceptual frameworks and case interpretations follow the original lecture by Prof. Eric Tsui.